

# Neutron Star

Deepak Kumar

August 5, 2022

## 1 Introduction

It has been a continuous endeavour over the years of curious human mind to understand the basic constituents of matter and their interactions. As per our current understanding, all the matter that we observe in nature and their interactions can be described by few elementary particles and four types of fundamental forces or interactions. The four fundamental forces in nature are 1) gravitational force, 2) weak force, 3) electromagnetic force and 4) strong force. Among them two forces, gravitational force and electromagnetic force, are long range forces and we experience them in day to day life. The gravitational force is the first fundamental force that was discovered, it is the one which is least understood at the fundamental level. Such force is responsible for the fall of the fruits from the tree to the motion of planets orbiting the stars and stars orbiting the galaxies etc. The most understood and common force is the electromagnetic force which manifests itself in the macroscopic world and is used in modern technology. On the atomic level, it is the electromagnetic interaction of the electrons in the atoms that eventually gives rise to different chemical properties of elements. On the other hand, the weak and strong forces are the short range forces so they are limited to the field of nuclear and particle physics. An example of the weak force is the  $\beta$  radiation of neutron where a free neutron decays to a proton, an electron and an antineutrino. Finally the strong interaction which is the strongest force among all forces binds the quarks to form nucleons and eventually the nucleons to form the nucleus. Apart from the gravity, the rest three fundamental interactions can be described by the gauge theories. The fundamental particles constitute three generations of quarks which can interact through strong, electromagnetic and weak interactions. The other set of fundamental particles, the three generation of leptons i.e. electron,  $\mu$ on and  $\tau$ on and the corresponding neutrinos ( $\nu_e$ ,  $\nu_\mu$  and  $\nu_\tau$ ) can not have strong interactions. Apart from this all of these particles interact with the higgs field. This essentially describes the standard model of particle physics.

## 2 Neutron star

Neutron Star (NS) is the remnant of the supernovae explosion. It is the second densest object in the universe after the black hole. The density of matter in the core of NS is few times nuclear saturation density ( $\rho_0 \sim 10^{14}$  gm cm<sup>3</sup>). At this ultra-high density one could expect matter with rich phases in the core. The phase of matter in the core is still unknown. The core, in principle, can support various possible exotic phases of Quantum Chromodynamics (QCD). While perturbative Quantum Chromodynamics (pQCD) predicts deconfined quark matter at large densities, their applicability is rather limited in the sense that these conclusions are applicable only to very large baryon densities i.e.  $\rho_B \geq 40\rho_0$  [1]. The most challenging region to study theoretically is, however, at intermediate densities i.e. few times nuclear matter saturation density which is actually relevant for the matter in the core of neutron stars. The first principle Lattice Quantum Chromodynamics (LQCD) calculation in this connection is also difficult due to

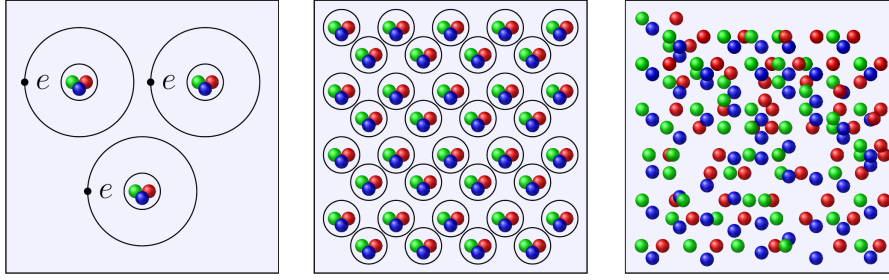


Figure 1: Schematic view of matter as increasing density from left to right (normal matter  $\rightarrow$  nuclear matter  $\rightarrow$  quark matter).

sign problem in lattice simulations at finite densities. At present such calculations are limited to low baryon densities only i.e.  $\mu_B/T \leq 3.5$  [2]. On the otherhand many effective models predict the possibilities of various exotic phases of quark matter at such intermediate density region. These include pion superfluidity [3–5], various color superconducting phases like 2-flavor color superconductivity [6–8], color flavor locked phase (CFL) [9], Larkin-Ovchinkov-Fulde-Ferrel (LOFF) [10, 11] phase, crystalline superconductivity phase etc. However, the signature of such phases in quark matter from the study of neutron stars have been rather challenging. The GW170817 [12] event explored the constraints on the Equation of State (EOS) using tidal deformability extracted from the phase of the gravitational waveforms during the late stage of inspiral merger [13–18]. Though not conclusive, it is quite possible that one or both the merging neutron stars could be hybrid stars *i.e.* with a core of quark matter or a mixed phase core of quark and hadronic matter [19, 20].

The NSs are the natural laboratories. They provide the low temperature and high density region of the QCD phase diagram Fig. ???. In the astrophysicist point of view, we observe the macroscopic properties of compact stars like mass, radius, tidal deformability, rotating frequency, moment of inertia, temperature etc. These macroscopic properties can be described from the microscopic nature of the matter inside the core of the star. It's a big question whether the matter inside the NS is nuclear matter, or quark matter, or a mixer of nuclear and quark matter. Let's come to the fundamental questions about the phases of matter at high densities coined in the previous paragraph. If we increase the baryon number density by squeezing the matter then after a level of squeezing the atoms or molecules desolve into the neutrons and protons where the neutron and proton emerge as the degrees of freedom of the matter, we call it nuclear matter (see the Fig. 1). If we continue the squeezing of matter further then we might reach another level where the neutrons and protons lost their identity, the matter becomes a soup of quarks, another degree of freedom of matter at this stage, we call it quark matter.

The typical mass of a canonical neutron star is about  $2M_\odot$  ( $M_\odot$  denotes the mass of sun) and radius is about 10 km. The pressure, to support the high mass star with small radius, should be large enough against the gravitational collapse. The pressure increases as we go from the surface to the center of the compact stars. We may expect different phases of matter at different radial distances. In Fig. 2 we display the cross-section view of neutron star. The inner core of NS is still unknown. We discuss the possible hadron-quark phase transition at ultra-high density in chapter Chapter ??. We consider nuclear matter described by Walecka type Relativistic Mean Field (RMF) model and quark matter described by Nambu–Jona-Lasinio (NJL) model. We establish hadron-quark phase transition using Gibbs construct mechanism.

Before going to discuss the Neutron Star Matter (NSM) in details, let's discuss the structure of compact stars. The structure of the compact stars can be found along with the EOS by the

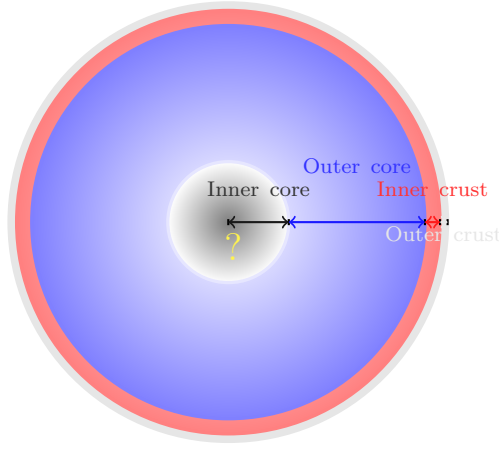


Figure 2: Schematic view of different layers in neutron stars.

Tolmann-Openheimer-Volkoff (TOV) equations.

$$p' = -(p + \epsilon) \left[ \frac{m + 4\pi r^3 p}{r(r - 2m)} \right] \quad (1)$$

$$m' = 4\pi r^2 \epsilon \quad (2)$$

where ‘prime’ denotes the derivative with respect to  $r$ . Eq. (1) is the hydrostatic equilibrium equation, where  $\epsilon$  and  $p$  are the energy density and pressure are the function of radial distance  $r$  from the center of the star. Eq. (2) is the mass balance equation.  $m$  is the total mass of the matter enclosed within the radius  $r$ . These equations are the coupled equations. These equations should be supplemented by EOS  $p = p(\epsilon)$ . These equations constitute a close system of equations to be solved for a given EOS to obtain mass-radius curve of compact stars. To solve these equations we have taken the following initial conditions.

$$m(r = 0) = 0 \quad \text{and} \quad p(r = 0) = p_0$$

where  $p_0$  is the central pressure inside the compact star.

Within the current observational status, it is difficult to distinguish between a canonical neutron star without a quark matter core from a hybrid star with a core of pure quark matter or a core of quark matter in a mixed phase with hadronic matter. This calls for exploring other observational signature to solve this “MASQUERADE” problem [21, 22]. In this context, it has been suggested that the study of the non-radial oscillation modes of NS can have the possibility of providing the compositional information regarding the matter in the interior of the NS. This includes the NS with a hyperon core [23–25], a quark core or a mixed phase core with quark and hadronic matter components [22, 26–31]. The pulsating equations that describe oscillations can be obtained by the perturbed Einstein field equations  $\delta G_{\alpha\beta} = 8\pi\delta T_{\alpha\beta}$  with  $G_{\alpha\beta} = R_{\alpha\beta} - \frac{1}{2}g_{\alpha\beta}R$  being the Einstein tensor. Linearizing these equations in the perturbation, the differential equations further fluid perturbation functions can be obtained as [32]

$$Q' - \frac{1}{c_s^2} [\omega^2 r^2 e^{\lambda-2\nu} Z + \nu' Q] + l(l+1)e^\lambda Z = 0 \quad (3)$$

$$Z' - 2\nu' Z + e^\lambda \frac{Q}{r^2} - \nu' \left( \frac{1}{c_e^2} - \frac{1}{c_s^2} \right) \left( Z + \nu' e^{-\lambda+2\nu} \frac{Q}{\omega^2 r^2} \right) = 0 \quad (4)$$

The two coupled first order differential equations for the perturbing functions  $Q(r, t)$  and  $Z(r, t)$ , Eqs.(3) and (4), are to be solved with appropriate boundary conditions at the center

and the surface. Near the center of the compact stars the behavior of the functions  $Q(r)$  and  $Z(r)$  are given by [26]

$$Q(r) = Cr^{l+1} \quad \text{and} \quad Z(r) = -Cr^l/l \quad (5)$$

where  $C$  is an arbitrary constant and  $l$  is the order of the oscillation. The other boundary condition is the vanishing of the Lagrangian perturbation pressure, *i.e.*  $\Delta p = 0$  at the stellar surface. Which gives

$$[\omega^2 r^2 e^{\lambda-2\nu} Z + \nu' Q]_{r=R} = 0. \quad (6)$$

Where the field functions  $\nu$  and  $\lambda$  are defined as

$$e^{-2\lambda} = 1 - \frac{2m}{r} \quad (7)$$

$$\nu' = \frac{m + 4\pi r^3 p}{r(r - 2m)} \quad (8)$$

The field function  $\nu$  can be found by integrating Eq.(8) from the center of the star to radial distance  $r$ . The integration constant is determined by satisfying the following

$$e^{2\nu(R)} = 1 - \frac{2M}{R}. \quad (9)$$

## References

- [1] Tyler Gorda et al. “Next-to-Next-to-Next-to-Leading Order Pressure of Cold Quark Matter: Leading Logarithm”. In: *Phys. Rev. Lett.* 121.20 (2018), p. 202701. DOI: 10.1103/PhysRevLett.121.202701. arXiv: 1807.04120 [hep-ph].
- [2] S. Borsányi et al. “Lattice QCD equation of state at finite chemical potential from an alternative expansion scheme”. In: *Phys. Rev. Lett.* 126.23 (2021), p. 232001. DOI: 10.1103/PhysRevLett.126.232001. arXiv: 2102.06660 [hep-lat].
- [3] D. T. Son and M. A. Stephanov. “QCD at Finite Isospin Density”. In: *Phys. Rev. Lett.* 86 (4 2001), pp. 592–595. DOI: 10.1103/PhysRevLett.86.592. URL: <https://link.aps.org/doi/10.1103/PhysRevLett.86.592>.
- [4] D. Ebert and K. G. Klimenko. “Pion condensation in electrically neutral cold matter with finite baryon density”. In: *Eur. Phys. J. C* 46 (2006), pp. 771–776. DOI: 10.1140/epjc/s2006-02527-5. arXiv: hep-ph/0510222.
- [5] A. Barducci et al. “A Calculation of the QCD phase diagram at finite temperature, and baryon and isospin chemical potentials”. In: *Phys. Rev. D* 69 (2004), p. 096004. DOI: 10.1103/PhysRevD.69.096004. arXiv: hep-ph/0402104.
- [6] Mark G. Alford, Krishna Rajagopal, and Frank Wilczek. “QCD at finite baryon density: Nucleon droplets and color superconductivity”. In: *Phys. Lett. B* 422 (1998), pp. 247–256. DOI: 10.1016/S0370-2693(98)00051-3. arXiv: hep-ph/9711395.
- [7] Amruta Mishra and Hiranmaya Mishra. “Chiral symmetry breaking, color superconductivity and color neutral quark matter: A Variational approach”. In: *Phys. Rev. D* 69 (2004), p. 014014. DOI: 10.1103/PhysRevD.69.014014. arXiv: hep-ph/0306105.
- [8] Aman Abhishek and Hiranmaya Mishra. “Chiral Symmetry Breaking, Color Superconductivity, and Equation of State for Magnetized Strange Quark Matter”. In: *Springer Proc. Phys.* 261 (2021). Ed. by Prafulla Kumar Behera et al., pp. 593–598. DOI: 10.1007/978-981-33-4408-2\_82.

- [9] Mark G. Alford, Krishna Rajagopal, and Frank Wilczek. “Color flavor locking and chiral symmetry breaking in high density QCD”. In: *Nucl. Phys. B* 537 (1999), pp. 443–458. DOI: 10.1016/S0550-3213(98)00668-3. arXiv: hep-ph/9804403.
- [10] Massimo Mannarelli, Krishna Rajagopal, and Rishi Sharma. “Testing the Ginzburg-Landau approximation for three-flavor crystalline color superconductivity”. In: *Phys. Rev. D* 73 (2006), p. 114012. DOI: 10.1103/PhysRevD.73.114012. arXiv: hep-ph/0603076.
- [11] Krishna Rajagopal and Rishi Sharma. “The Crystallography of Three-Flavor Quark Matter”. In: *Phys. Rev. D* 74 (2006), p. 094019. DOI: 10.1103/PhysRevD.74.094019. arXiv: hep-ph/0605316.
- [12] B. P. Abbott et al. “GW170817: Measurements of neutron star radii and equation of state”. In: *Phys. Rev. Lett.* 121.16 (2018), p. 161101. DOI: 10.1103/PhysRevLett.121.161101. arXiv: 1805.11581 [gr-qc].
- [13] David Radice et al. “GW170817: Joint Constraint on the Neutron Star Equation of State from Multimessenger Observations”. In: *The Astrophysical Journal* 852.2 (2018), p. L29. DOI: 10.3847/2041-8213/aaa402. URL: <https://doi.org/10.3847/2041-8213/aaa402>.
- [14] Tuhin Malik et al. “GW170817: constraining the nuclear matter equation of state from the neutron star tidal deformability”. In: *Phys. Rev. C* 98.3 (2018), p. 035804. DOI: 10.1103/PhysRevC.98.035804. arXiv: 1805.11963 [nucl-th].
- [15] Cheng-Ming Li et al. “Constraints on the hybrid equation of state with a crossover hadron-quark phase transition in the light of GW170817”. In: *Phys. Rev. D* 98.8 (2018), p. 083013. DOI: 10.1103/PhysRevD.98.083013. arXiv: 1808.02601 [nucl-th].
- [16] Jinniu Hu et al. “Effects of symmetry energy on the radius and tidal deformability of neutron stars in the relativistic mean-field model”. In: *PTEP* 2020.4 (2020), p. 043D01. DOI: 10.1093/ptep/ptaa016. arXiv: 2002.00562 [nucl-th].
- [17] Soumi De et al. “Tidal Deformabilities and Radii of Neutron Stars from the Observation of GW170817”. In: *Phys. Rev. Lett.* 121.9 (2018). [Erratum: *Phys.Rev.Lett.* 121, 259902 (2018)], p. 091102. DOI: 10.1103/PhysRevLett.121.091102. arXiv: 1804.08583 [astro-ph.HE].
- [18] Katerina Chatziioannou, Carl-Johan Haster, and Aaron Zimmerman. “Measuring the neutron star tidal deformability with equation-of-state-independent relations and gravitational waves”. In: *Phys. Rev. D* 97.10 (2018), p. 104036. DOI: 10.1103/PhysRevD.97.104036. arXiv: 1804.03221 [gr-qc].
- [19] Vasileios Paschalidis et al. “Implications from GW170817 and I-Love-Q relations for relativistic hybrid stars”. In: *Phys. Rev. D* 97.8 (2018), p. 084038. DOI: 10.1103/PhysRevD.97.084038. arXiv: 1712.00451 [astro-ph.HE].
- [20] Rana Nandi and Prasanta Char. “Hybrid stars in the light of GW170817”. In: *Astrophys. J.* 857.1 (2018), p. 12. DOI: 10.3847/1538-4357/aab78c. arXiv: 1712.08094 [astro-ph.HE].
- [21] Mark Alford et al. “Hybrid stars that masquerade as neutron stars”. In: *Astrophys. J.* 629 (2005), pp. 969–978. DOI: 10.1086/430902. arXiv: nucl-th/0411016.
- [22] Wei Wei et al. “Lifting the Veil on Quark Matter in Compact Stars with Core  $g$ -mode Oscillations”. In: *Astrophys. J.* 904.2 (2020), p. 187. DOI: 10.3847/1538-4357/abbe02. arXiv: 1811.11377 [nucl-th].

- [23] V. A. Dommes and M. E. Gusakov. “Oscillations of superfluid hyperon stars: decoupling scheme and g-modes”. In: *Mon. Not. Roy. Astron. Soc.* 455.3 (2016), pp. 2852–2870. DOI: 10.1093/mnras/stv2408. arXiv: 1512.04900 [astro-ph.SR].
- [24] Hang Yu and Nevin N. Weinberg. “Resonant tidal excitation of superfluid neutron stars in coalescing binaries”. In: *Mon. Not. Roy. Astron. Soc.* 464.3 (2017), pp. 2622–2637. DOI: 10.1093/mnras/stw2552. arXiv: 1610.00745 [astro-ph.HE].
- [25] Bikram Keshari Pradhan and Debarati Chatterjee. “Effect of hyperons on f-mode oscillations in Neutron Stars”. In: *Phys. Rev. C* 103.3 (2021), p. 035810. DOI: 10.1103/PhysRevC.103.035810. arXiv: 2011.02204 [astro-ph.HE].
- [26] Hajime Sotani et al. “Signatures of hadron-quark mixed phase in gravitational waves”. In: *Phys. Rev. D* 83 (2011), p. 024014. DOI: 10.1103/PhysRevD.83.024014. arXiv: 1012.4042 [astro-ph.HE].
- [27] C. V. Flores and G. Lugones. “Discriminating hadronic and quark stars through gravitational waves of fluid pulsation modes”. In: *Class. Quant. Grav.* 31 (2014), p. 155002. DOI: 10.1088/0264-9381/31/15/155002. arXiv: 1310.0554 [astro-ph.HE].
- [28] Alessandro Brillante and Igor N. Mishustin. “Radial oscillations of neutral and charged hybrid stars”. In: *EPL* 105.3 (2014), p. 39001. DOI: 10.1209/0295-5075/105/39001. arXiv: 1401.7915 [astro-ph.SR].
- [29] Ignacio F. Ranea-Sandoval et al. “Oscillation modes of hybrid stars within the relativistic Cowling approximation”. In: *JCAP* 12 (2018), p. 031. DOI: 10.1088/1475-7516/2018/12/031. arXiv: 1807.02166 [astro-ph.HE].
- [30] M. C. Rodriguez et al. “Hybrid stars with sequential phase transitions: the emergence of the  $g_2$  mode”. In: *JCAP* 02 (2021), p. 009. DOI: 10.1088/1475-7516/2021/02/009. arXiv: 2009.03769 [astro-ph.HE].
- [31] Shu Yan Lau and Kent Yagi. “Probing hybrid stars with gravitational waves via interfacial modes”. In: *Phys. Rev. D* 103.6 (2021), p. 063015. DOI: 10.1103/PhysRevD.103.063015. arXiv: 2012.13000 [astro-ph.HE].
- [32] Hajime Sotani. “Neutron star asteroseismology and nuclear saturation parameter”. In: *Phys. Rev. D* 103.12 (2021), p. 123015. DOI: 10.1103/PhysRevD.103.123015. arXiv: 2105.13212 [astro-ph.HE].